



Underwater Acoustics Assignment

By

Hyejoo Kwon, Sadhasivam Mukesh

Submitted to: M. Saillard

Universite de Toulon

January 29, 2025



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Introduction

In this report, we delve into the trajectory of acoustic waves in the deep sea. Due to the limitations of electromagnetic waves in water, acoustic waves are vital for communication and navigation. Acoustic waves can propagate over long distances through water and are influenced by factors such as water depth, pressure, salinity, and temperature. In this report, we specifically focus on the properties of wave trajectory with respect to water depth, wave angle and wave velocity. To understand the trajectory of sound waves, we use Matlab and Python to plot the results. By using these tools, this report aims to provide valuable insights into the relationship between wave velocity, wave angle, and water depth.

1 Homework 1

Arctic Ocean

Exercise 1

Let us consider a transmitter at depth z_S . What are the geometrical characteristics of the trajectory of a ray transmitted with an initial angle θ_0 toward the sea bottom (measured from the horizontal axis)? Plot this ray for $\theta_0 = 2^\circ$ and $z_S = 500$ m with the help of your ray tracing software.

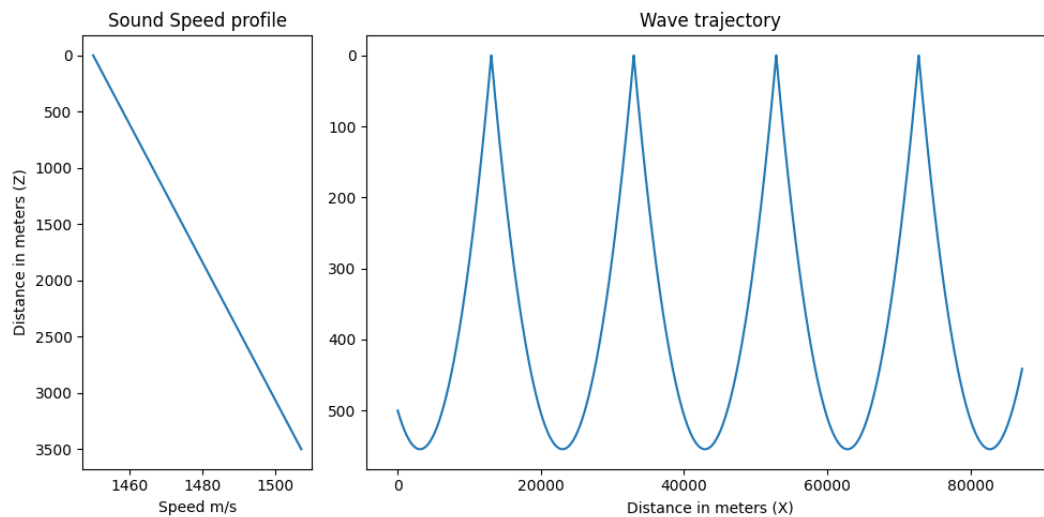


Figure 1: Lift Speed

Before analyzing the geometric characteristics of the wave's trajectory, we first examine the sound speed profile. From the sound velocity graph, we observe that the sound speed gradient is linear. According to the properties of sound waves, if the velocity gradient is constant, the wave's trajectory forms an elliptical or circular shape. We expected the



trajectory to resemble an ellipse because of the limitation of the depth of the water and small initial angle. As you see the figure (1), we found that the shape of the wave trajectory is ellipse. Specifically, the waves initially moves downward but gradually bends toward a horizontal direction. This pattern results from the interaction between the sound speed variation and Snell's law as following :

$$\frac{\sin \theta}{c(z)} = \text{constant}$$

where θ is the propagation angle, and $c(z)$ is the sound speed at depth z . As the wave goes deeper and the sound speed $c(z)$ increases, $\sin \theta$ decreases to maintain the constant value. This means that the angle of the wave becomes smaller, causing the wave to bend upward. When the wave reaches the surface, it reflects due to the large impedance difference between water and air. This is why the trajectory of wave repeats the bending and reflection pattern. As a result, this causes the wave to bend upward around 600 meters and heads back toward the surface and reflects.

Exercise 2

We wish to establish a communication between two hydrophones located at $(0, 0, z_S)$ and $(x, 0, z_S)$, respectively. Express, as a function of c_0 , γ , x , and z_S an emission angle θ_0 generating a ray whose path passes through both hydrophones. Check the result using ray tracing software for $x = 20$ km and $z_S = 500$ m (to the nearest 0.1° for θ_0). Explore other possibilities. Suggest a solution for operational conditions and explain why.

The equation of the emission angle is derived based on Snell's law and total displacement of x . Since the sound speed increases linearly with depth, we can calculate the total horizontal distance x_{total} . When the sound wave reflects multiple times between the surface and the seabed, the total horizontal distance can be represented as the product of the number of reflections n and the horizontal distance Δx during one reflection. Here, n , the number of reflections, can be determined from the radius of the wave trajectory. The radius R is defined as:

$$R = \frac{c_0 + \gamma_0 z_s}{\sin \theta_s \cdot \left| \frac{dc}{dz} \right|}$$

The horizontal distance for one up/down path is given by:

$$x_{\text{one}} = 2 \cdot R \cdot \sin \theta_s$$



If the sound wave have n reflections, the total horizontal distance x_{total} is:

$$x_{\text{total}} = n \cdot x_{\text{one}} = 2 \cdot n \cdot R \cdot \sin \theta_s$$

Substituting R into the above equation, The total horizontal distance is expressed as:

$$x_{\text{total}} = 2 \cdot n \cdot \frac{c_0 + \gamma_0 z_s}{\sin \theta_s \cdot \left| \frac{dc}{dz} \right|} \cdot \sin \theta_s$$

Simplifying by canceling out $\sin \theta_s$:

$$n = \frac{x_{\text{total}} \cdot \left| \frac{dc}{dz} \right|}{2 \cdot c_0 + \gamma_0 z_s}$$

In order to derive $\sin(\theta_s)$, rearrange the snell's law equation by substituting the above equations as follows:

$$\sin(\theta_s) = \frac{x_{\text{total}} \cdot \gamma_0}{2 \cdot n \cdot (c_0 + \gamma_0 z_s)}.$$

Finally, the function of θ_s with respect to c_0, γ_0, z_s, x is determined by taking the arcsine:

$$\theta_s = \arcsin \left(\frac{x_{\text{total}} \cdot \gamma_0}{2 \cdot n \cdot (c_0 + \gamma_0 z_s)} \right).$$

We only consider a single value of $n = 1$ to determine the optimal angle under the given operational conditions. The reason for not considering other values of n is to ensure communication stability between the two hydrophones. As the number of reflections increases, the likelihood of instability also increases. Additionally, if we want the wave to reflect more than once, the emission angle must be regulated to exceed 10 degrees, which would result in a sharper and more rapid wave trajectory. Consequently, according to the equation derived above, the emission angle is calculated to be approximately 6.2 degrees. Considering potential calculation errors, we take an error margin of ± 0.2 with an interval of 0.01. As a result we found the optimal value with an error less than 1, which is 6.38 degrees.

Delving into other possibilities, we considered numerical solutions. However, we encountered limitations when implementing the numerical method in the code. To calculate a wide range from 0 to ± 90 degrees with a small step size, the number of iterations had to be significantly increased, which required a lot of time to execute the code. For this reason, a code-based numerical solution was not selected.

Mediterranean Sea

Exercise 3

In the Mediterranean Sea, we observe a decrease in sound velocity over 700 m with a gradient of -0.026 s^{-1} before returning to the Arctic Ocean-like behavior.

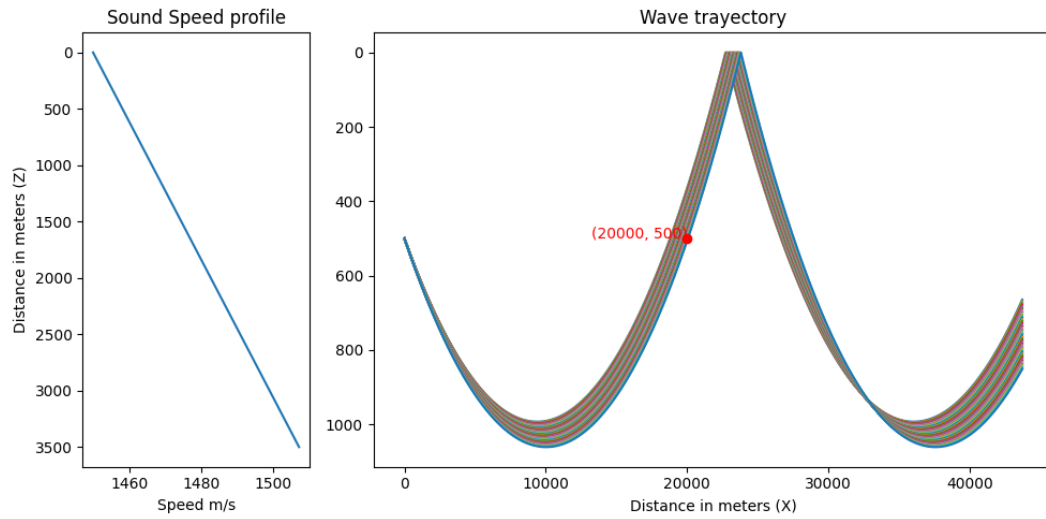


Figure 2: Wave trajectory through (20000,500) with margin

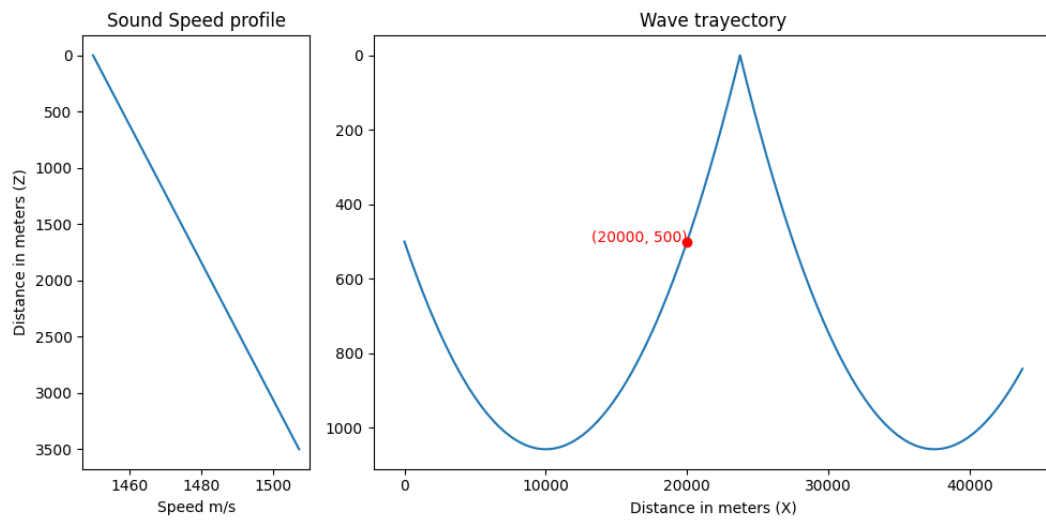


Figure 3: Wave trajectory through (20000,500) with 6.38 degree



For communication with a receiver 10 km away at a depth of 1000 m, what z_S and emission angle θ_0 would you suggest? Do we have the choice for z_S ? Suggest a strategy for finding a solution rapidly under operational conditions.

We conclude that placing the transmitter at a depth of around 700 meters is the most suitable choice. This is because a sound channel is likely to form at the point where the gradient of sound velocity changes. Specifically, when the sound velocity decreases, sound waves naturally refract downward. Conversely, when the sound velocity increases, the waves refract upward. At the point where the gradient of sound velocity changes its sign, these two processes repeat, causing the sound waves to oscillate above and below the middle depth of around 700 meters. We assume that this phenomenon allows the waves to travel longer distances with less energy loss and enhances their stability.

Therefore, we aim to find the optimal angle that intersects the two points (0,700) and (10,000,1,000) using the method mentioned above. The emission angle was calculated to be about 4.8° . To determine the optimal point, we considered an error margin of $\pm 0.1^\circ$ with a 0.01° interval. The error was calculated as follows:

$$\text{Error in } x : |x_{\text{predict}} - x|, \quad \text{Error in } z : |z_{\text{predict}} - z|$$

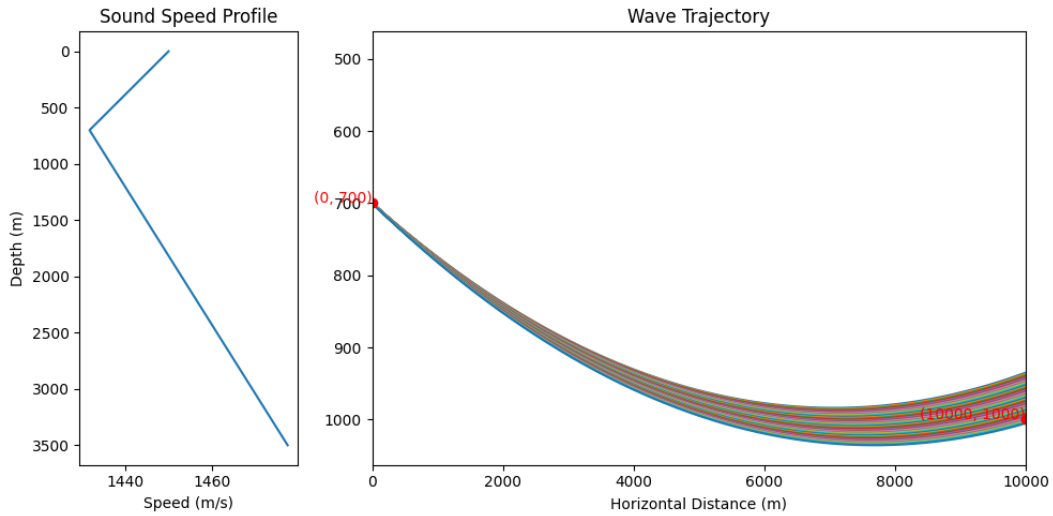


Figure 4: Wave trajectory through (10000,1000) with margine

As shown in the figure 5, most trajectories are close to the point (10,000,1,000). The most accurate point, with an error of less than 2, was found to be at 4.97° .

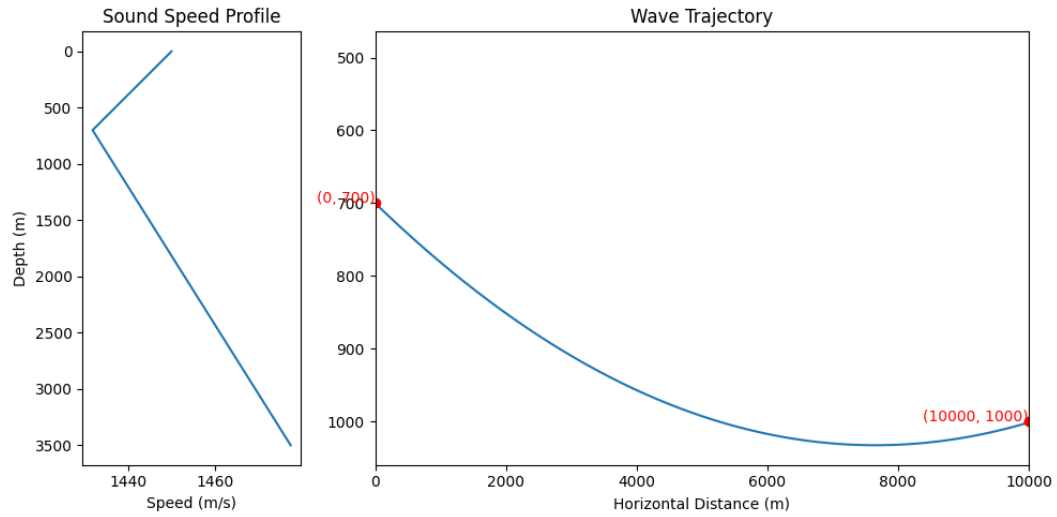


Figure 5: Wave trajectory through (10000,1000) with 4.97 degree

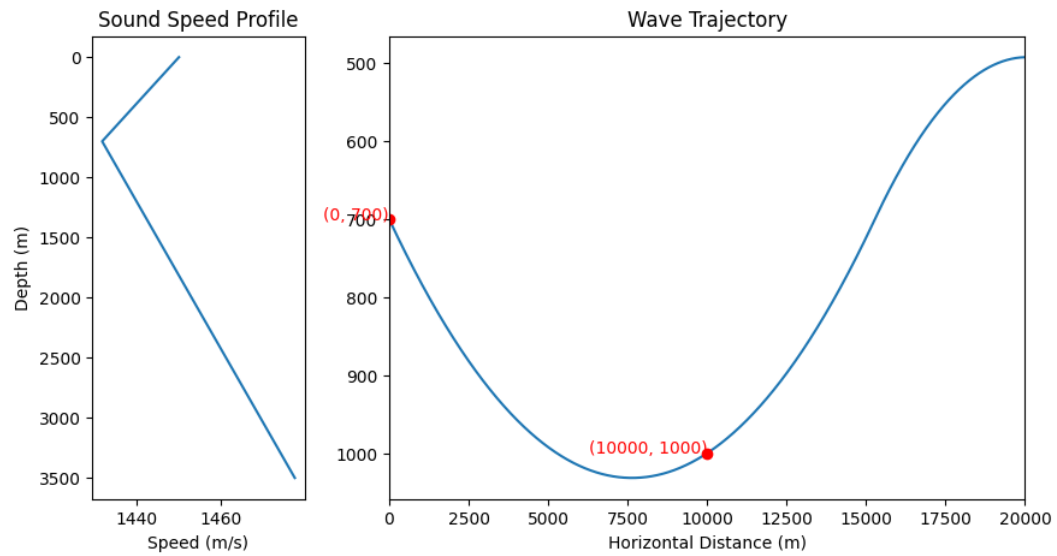


Figure 6: Wave trajectory through (10000,1000) with 4.97 degree and longer x range



Conclusion

We studied how acoustic waves move in the Arctic Ocean and Mediterranean Sea to improve underwater communication. By using simple calculations and ray tracing, we found the best emission angles (6.38° and 4.97°) and the right transmitter depth, such as 700 meters in the Mediterranean, where sound waves travel better because of changes in sound speed. While we thought about using numerical methods, they were too slow and complicated, so we used simpler approaches instead. This study helps us better understand how sound travels underwater for communication and navigation.



2 Homework 2

Plot Received Signals and Rough Depth Estimation

The assignment requires plotting the received signals from nine receivers placed at depths of $25n$ meters ($1 \leq n \leq 9$) over 6.4 seconds. The signal has a sampling time step of $dt = 2 \times 10^{-4}$ s, and the speed of sound is assumed to be $c = 1500$ m/s. The goal is to estimate the depth of the water column and the source location based on the time delays.

Solution

The plot of the received signals is shown below:

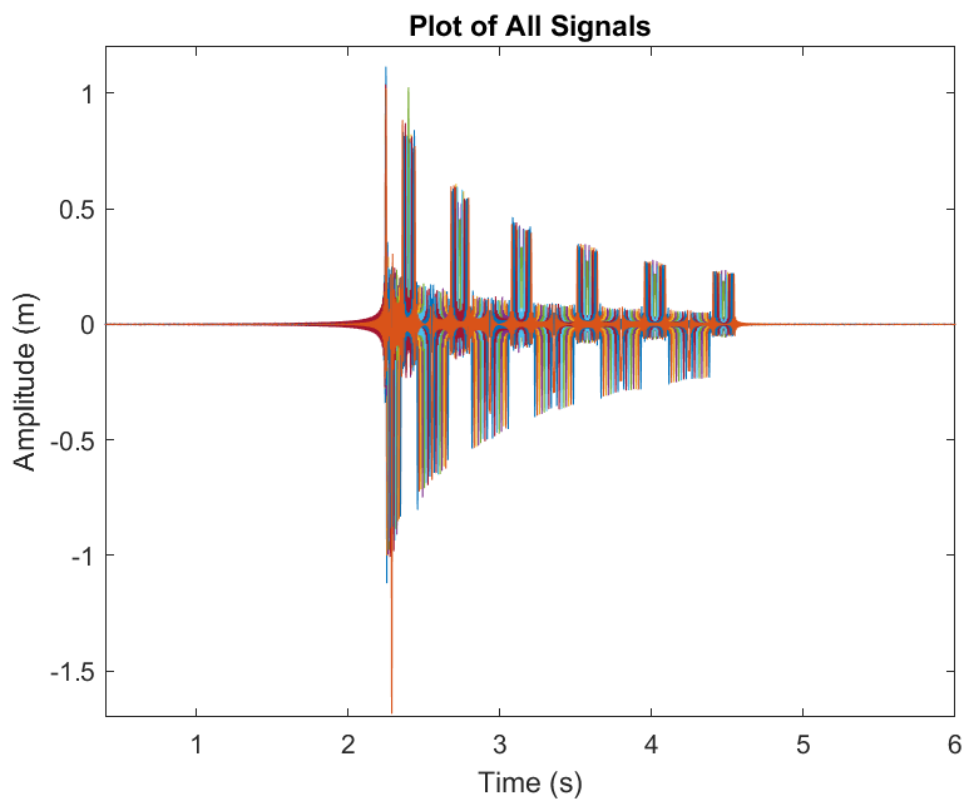


Figure 7: Plot of the received signals from 9 receivers.

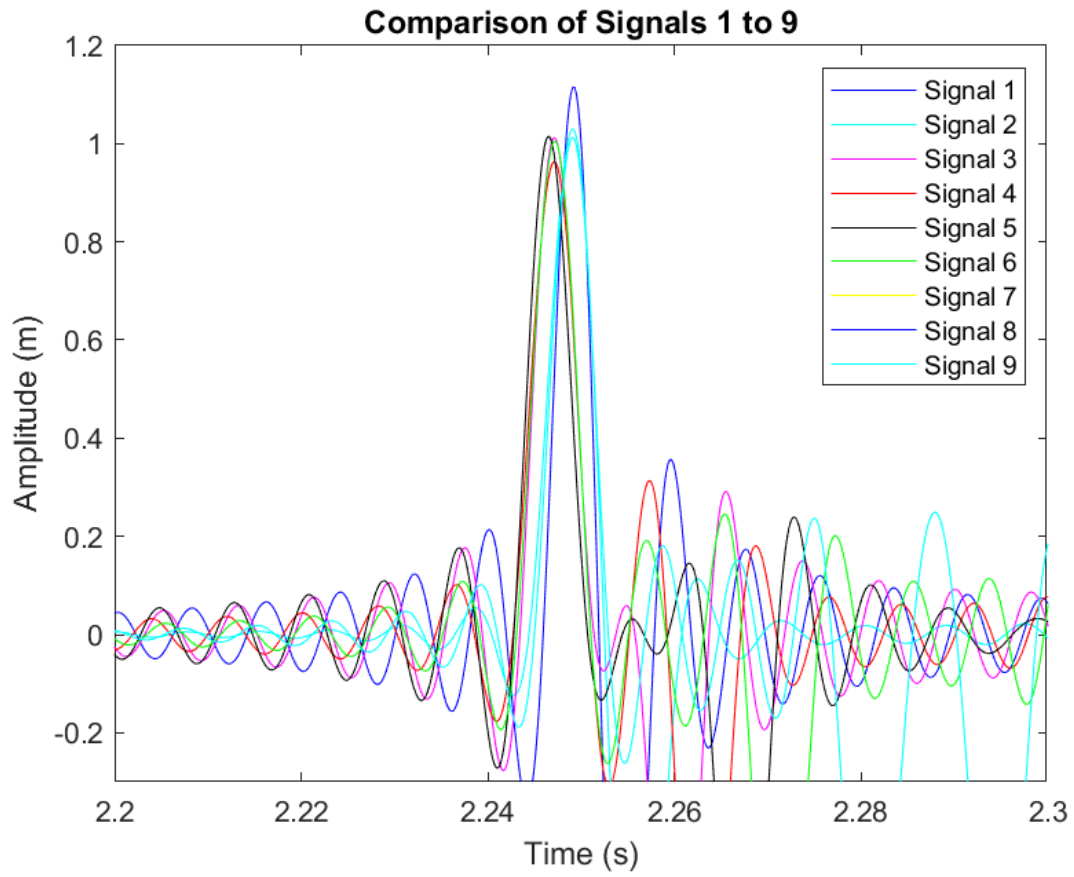


Figure 8: Plot signals around first peak

Estimating Depth h and Source Location

We observe the time delays between the receivers' signals which show that they have different arrival times.

Time Delay Analysis

In this step, we plot the received signals from the '.mat' file and identify the time of the first signal peak at each receiver. This allows us to observe the time delays, which help estimate the source depth.

Receiver Depth (m)	Time of First Peak (s)
25	2.2492
50	2.2490
75	2.2470
100	2.2470
125	2.2464
150	2.2472
175	2.2476
200	2.2490
225	2.2504

Table 1: Arrival times of the first peak for each receiver.

Source Distance Estimation Using Pythagorean Calculation

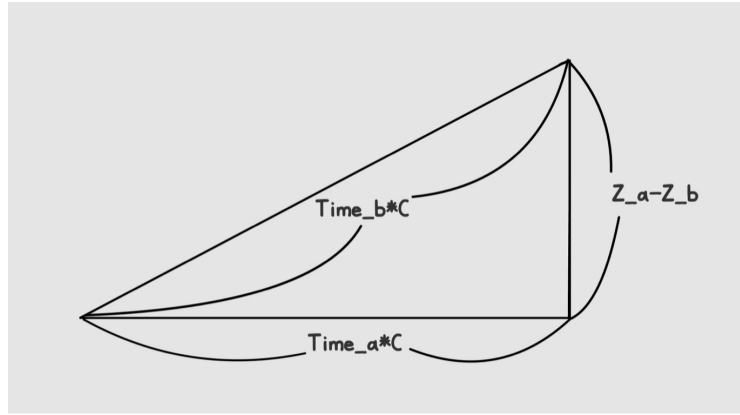


Figure 9: Pythagorean Calculation

First of all, since Receiver 5 has the shortest arrival time, we assume that it is located on the same horizontal line as the transmitter. Therefore, we calculated the distance between the transmitter and the receiver based on Receiver 1 and Receiver 5.

We use the Pythagorean theorem to calculate the horizontal distance from the transmitter to the receiver, based on the time delays between two receivers.

Given:

- Speed of sound $c = 1500 \text{ m/s}$
- Time difference between receiver at $z_a = 125 \text{ m}$ and $z_b = 25 \text{ m}$: $\Delta t = 2.2492 - 2.2464 = 0.0028 \text{ s}$
- Vertical distance $\Delta z = 100 \text{ m}$

The relationship is given by:

$$(t_b - t_a) \cdot c = \sqrt{(t_a \cdot c)^2 + (\Delta z)^2}$$



Rearranging and solving:

$$x_s = \sqrt{(c \cdot \Delta t)^2 - (\Delta z)^2}$$

Substituting the values as follows :

$$x_s = \sqrt{(1500 \cdot 0.0028)^2 - (100)^2} \approx 960 \text{ m}$$

Thus, the shortest horizontal distance between the transmitter and the receiver, using the formula above, is $x_s \approx 960 \text{ m}$.

Determine the depth of transmitter h

Since we calculate the distance between the transmitter and receiver, we can also derive a formula to find the depth of the transmitter. The signal is sent from the transmitter, reflects off the seafloor, and reaches the receiver. We assume that the signal path forms a triangle, so we can use the Pythagorean theorem to calculate the depth.

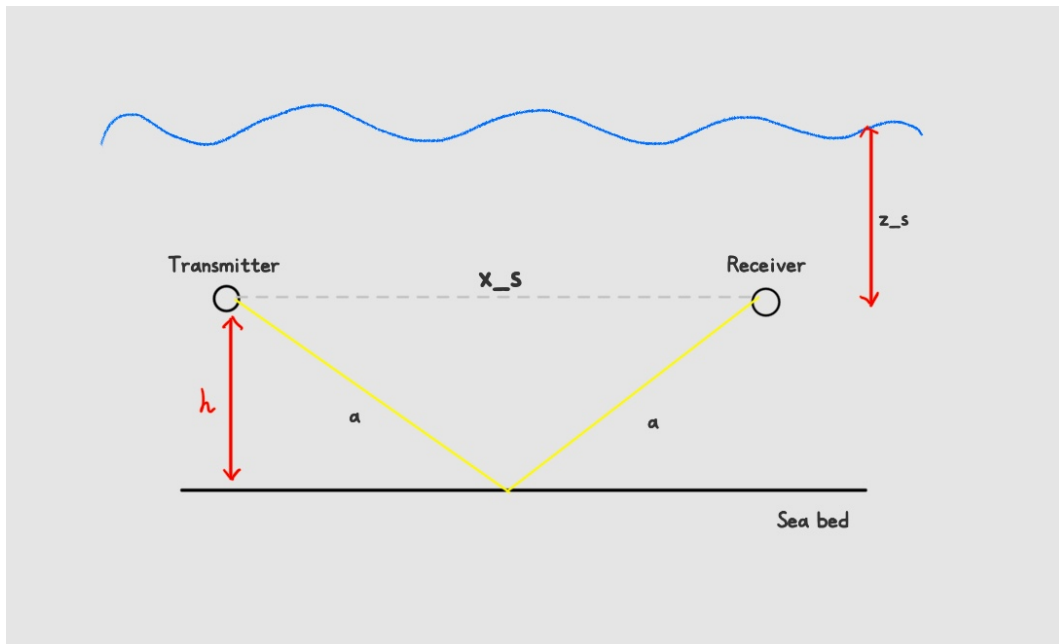


Figure 10: Wave path by the shortest receiver

- x_s : The horizontal distance between the transmitter and receiver : 960m
- δt : The time delay between first peak and first reflection peak of receiver 5 : $2.314472.24647=0.068\text{s}$
- c : The speed of sound in water : 1500m/s.
- h : The depth of the seafloor.
- z_s : The distance between water surface and receiver : 125m



Since the signal reflects before reaching the receiver, we need to consider the actual path the signal travels:

$$2a - x_s = \delta t \cdot c$$

$$a = \frac{x_s}{2} + \frac{\delta t \cdot c}{2}$$

Since the signal's path forms a right triangle, we use the Pythagorean theorem:

$$a^2 = h^2 + \left(\frac{x_s}{2}\right)^2$$

Substituting a from the previous step:

$$\left(\frac{x_s}{2} + \frac{\delta t \cdot c}{2}\right)^2 = h^2 + \left(\frac{x_s}{2}\right)^2$$

Rearranging:

$$h^2 = \frac{\delta t \cdot c \cdot x_s}{2} + \left(\frac{\delta t \cdot c}{2}\right)^2$$

Taking the square root on both sides and add the distance between receiver and water surface:

$$h_{\text{total}} = \sqrt{\frac{\delta t \cdot c \cdot x_s}{2} + \left(\frac{\delta t \cdot c}{2}\right)^2} + z_s$$

Substituting the values, the total depth is as follows :

$$\mathbf{h_{total} \approx 352.07 \text{ m}}$$

Problem 2: Backpropagation Simulation for Improved Accuracy

To improve the accuracy of the source location, we perform a backpropagation simulation by transmitting time-reversed signals.

Mathematical Formulation for Backpropagation

The backpropagation process involves the convolution of the time-reversed signals $s_i(t)$ with the Green's function $G(x_t, z_t, x_s, z_s, t)$:

$$s_i(t) * G(x_t, z_t, x_s, z_s, t) = - \sum_j \epsilon_j \frac{s(t - r_j/c)}{4\pi r_j}$$



where:

- r_j is the distance from the source to the receiver after multiple reflections.
- c is the speed of sound.
- ϵ_j accounts for phase inversion from reflections.

Simulation of Backpropagated signal

The simulation scans a grid from 940 m to 1050 m on the x-axis, divided into 20 sections, and from 110 m to 130 m in depth, also divided into 20 sections.

The first heatmap shows the most likely location of the transmitter, which is around (990 m, 121 m). However due to the big window size, we realize that the accuracy of this result is vague to identify the coordinate of the source. Therefore we plot one more heatmap with 30 windows to find more precise location. As a result, we found that the optimal coordinate of the sound source is located at (985,121).

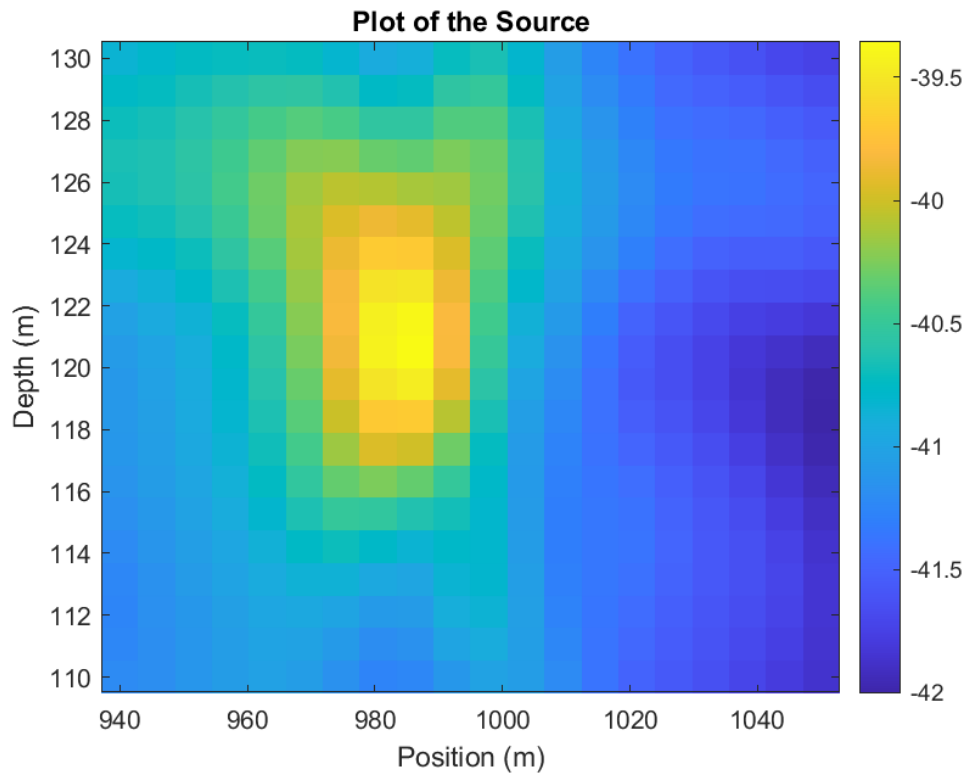


Figure 11: heatmap from grid scan of source location.

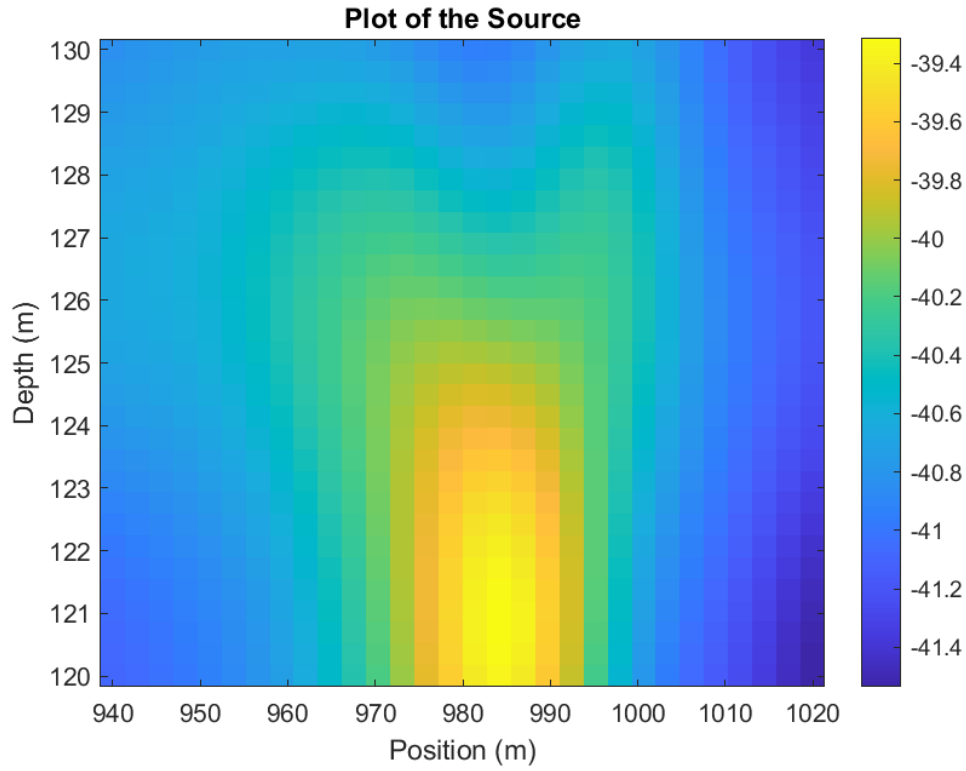


Figure 12: heatmap of source location with more windows.

Comparison with Provided Simulation Results

The reference simulation revealed a source location at approximately $x = 985$, $z = 121$. Compared to estimated x and z coordinate we found above, there is a slight variation in depth due to measurement inaccuracies.

Conclusions

Through time delay analysis and backpropagation simulation, we successfully estimated the source location in the water column. The analytical solution provided an initial estimate, while the experimental backpropagation method improved the accuracy.